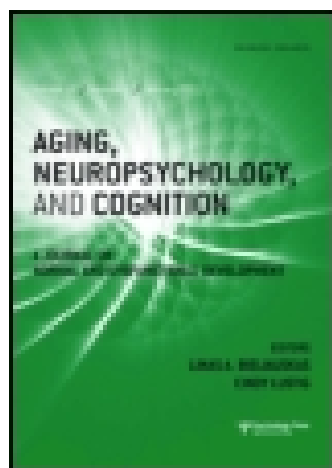


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Age-specific differences of dual n-back training

Tiina Salminen^a, Peter Frensch^a, Tilo Strobach^a & Torsten
Schubert^a

^a Department of Psychology, Humboldt-Universität zu Berlin,
Berlin, Germany

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Age-specific differences of dual *n*-back training

Tiina Salminen ^{*}, Peter Frensch, Tilo Strobach and Torsten Schubert

Department of Psychology, Humboldt-Universität zu Berlin, Berlin, Germany

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Age-related decline in executive functions can be decisive in performing everyday tasks autonomously. Working memory (WM) is closely related to executive functions, and training of WM has yielded evidence toward cognitive plasticity in older adults. The training effects often transfer to untrained tasks and functions. These effects have mostly been shown in processes such as WM and attention, whereas studies investigating transfer to executive functions have been scarce. We trained older adults aged 57–73 years in a WM training task that was reported to be effective in producing transfer in young adults. The training intervention consisted of a dual *n*-back task including independently processed auditory and visual *n*-back tasks. We investigated transfer to tasks engaging executive functions, and compared the effects in older adults to those reported in young adults. We found that both training groups improved in the training task. Although the training effect in older adults was smaller than the training effect in young adults, the older adults still showed a notable improvement so that after training they performed on the same level as young adults without training. The older adults also showed transfer to an untrained WM updating task, a result that was in accordance with the findings in young adults; other transfer effects in older adults were lacking. We conclude that although transfer effects were scarce, the present study provides encouraging evidence toward the possibilities to compensate for age-related decline in executive functions by a WM training intervention.

Keywords: working memory training; executive functions; cognitive plasticity; age-related differences; older adults; dual *n*-back

Introduction

Aging leads to various impairments in cognitive performance. Several studies have shown that these cognitive impairments are observable in tasks measuring functions such as processing speed (Salthouse, 1996), episodic memory (Nyberg, Bäckman, Erngrund, Olofsson, & Nilsson, 1996), and working memory (WM) (Kramer, Hahn, & Gopher, 1999; Park et al., 2002). Further, executive control functions are impaired with age (Hein & Schubert, 2004; Raz, 2000; West, 1996). According to a definition by Miyake et al. (2000), executive control functions comprise the processes of shifting (between tasks or mental sets), updating (of information in WM), and inhibition (of prepotent responses). As these functions play a crucial role in many everyday tasks including planning, reasoning, and problem solving, investigating the possibilities to overcome age-related decline can provide insights into preserving autonomy and abilities to function in everyday life at an older age.

*Corresponding author. Email: tiina.salminen@hu-berlin.de

Present address for Tilo Strobach is Medical School Hamburg, Hamburg, Germany

Using training interventions to compensate for age-related cognitive deficits has been of interest to researchers for several decades by now (e.g., Baltes & Willis, 1982; Kramer, Larish, & Strayer, 1995; Labouvie-Vief & Gonda, 1976). Earlier training studies often aimed to practice mnemonic strategies. In strategy-based training regimes participants simply have to learn a new strategy to improve their memory performance in a given task. However, the improvements achieved through strategy-based training are frequently not transferred to new tasks with new stimuli (Brooks, Friedman, & Yesavage, 1993; Mohs et al., 1998; Oswald, Rupperecht, Gunzelmann, & Tritt, 1996).

Over the last decade there has been an increase in interest toward process-based training, as shown by a large number of emerging studies (Kelly et al., 2014; Lustig, Shah, Seidler, & Reuter-Lorenz, 2009). Process-based training paradigms are designed to improve some cognitive process in general, without providing the participants with explicit strategies to perform the task. Since the aim of this type of training is to improve a cognitive function independent of the trained material, one would expect observable improvements also in untrained transfer tasks engaging the trained process (Schubert, Strobach, & Karbach, 2014).

One example of process-based training is WM training. Previous training studies with older adults have shown that WM processes can be improved still at an older age (for meta-analyses and reviews, see Au et al., 2014; Karbach & Verhaeghen, 2014; Kueider, Parisi, Gross, & Rebok, 2012). Yet, findings of transfer to untrained tasks and functions have been inconsistent. Some studies have reported improvements in untrained but structurally similar tasks (i.e., similar to the training task), and these effects are typically referred to as “near transfer” (e.g., Karbach & Kray, 2009). For example, Li et al. (2008) reported near transfer to a spatial 3-back task and numerical 2- and 3-back tasks following training on a spatial 2-back task and on a task requiring mental spatial shifting and updating of WM contents. This study, however, demonstrated no improvements in untrained and structurally dissimilar tasks (i.e., dissimilar to the training task), that is, “far transfer,” whereas other studies were able to do so. Richmond, Morrison, Chein, and Olson (2011), for instance, found after training on a spatial complex WM span task and on a verbal complex WM span task near transfer to a reading span task and far transfer to a test of verbal learning and memory, but no transfer to fluid intelligence or attention. Both near and far transfer were also found in a study by Carretti, Borella, Zavagnin, and De Beni (2012), who observed improvements in a verbal WM updating task as well as in language comprehension after verbal WM span training. A study by Stepankova et al. (2014) showed near transfer after verbal *n*-back training to untrained measures of WM and far transfer to visuospatial (VS) skills.

There are only a few studies that have systematically investigated transfer effects from WM training to multiple executive functions. For example, Dahlin, Stigsdotter Neely, Larsson, Bäckman, and Nyberg (2008) trained older adults on seven different WM updating tasks (each task requiring the encoding of different kinds of stimuli: numbers, letter, colors, or spatial locations) for 15 sessions over five weeks. Their measure for near transfer was a 3-back task and for far transfer the Stroop task to tap the executive function of inhibition. The authors did not observe transfer to either the 3-back or the Stroop task. Brehmer, Westerberg, and Bäckman (2012) investigated transfer effects from spatial and verbal WM training to untrained WM measures as well as to an attention task (PASAT) and the Stroop task. Near transfer was found to WM span tasks and far transfer to PASAT but not to the Stroop task. Finally, in a study by Zinke et al. (2014) older participants trained on three separate tasks (visual WM, auditory WM, and executive control) for nine sessions spanning over three weeks. They observed near transfer to a verbal WM task and

far transfer to a task measuring fluid intelligence, but no transfer to a Stroop task. Thus, there seems to be a lack of studies that have systematically investigated the possibilities of compensating for age-related decline in several executive functions via WM training (see also Dahlin, Nyberg, Bäckman, & Stigsdotter Neely, 2008; Li et al., 2008; for effects in WM updating). This is somewhat surprising, since 1) impairments in executive functions are prominent in old age and these shortcomings can crucially deteriorate abilities to function in everyday life (Raz, 2000; Salthouse, Atkinson, & Berish, 2003), and 2) executive functions are strongly associated with WM processes and therefore training of WM could potentially have an effect on executive functions, as well (Baddeley, 1996a, 1996b; Miyake & Friedman, 2012; Miyake et al., 2000).

Thus far, WM training paradigms for older adults have exclusively tapped either the visual or the auditory modality. In some studies, both modalities have been trained, but separately. However, there is ample evidence that adult age differences increase as a function of task complexity and the inclusion of a variety of process types (Salthouse, 1992; Voelcker-Rehage, Stronge, & Alberts, 2006). One possibility to increase task complexity and the number of tapped processes is to include concurrent input from two different modalities and tasks (Jaeggi, Buschkuhl, Jonides, & Perrig, 2008). An example of such a task, the dual *n*-back task, consists of separate VS and auditory-verbal (AV) *n*-back tasks, requiring the participants to monitor and update in their WM the contents from two different inputs (Buschkuhl, Jaeggi, Kobel, & Perrig, 2007). The VS and the AV tasks are independent of each other although presented synchronously. Consequently, the dual *n*-back task engages executive functions with a dual-task component of coordinating and controlling different task streams (Strobach, Salminen, Karbach, & Schubert, 2014). In order to suppress the development of task-specific strategies (and thus allowing the intervention to be rather process-based), to maximize the efficiency of training, and to trigger cognitive plasticity, the dual *n*-back training task has been created as an adaptive one: this means that the level of *n* in each trial is adapted according to the participants' performance (Jaeggi, Buschkuhl, et al., 2008; Klingberg, 2010).

In a study comparing age effects in single *n*-back performance (VS or AV *n*-back task) and dual *n*-back performance (VS and AV *n*-back task) without training, Jaeggi, Schmid, Buschkuhl, and Perrig et al. (2008) showed that the age effects were specifically prominent in the dual *n*-back version. That is, the older participants showed a significantly impaired performance in the dual *n*-back but not in the single *n*-back tasks, compared with the young adults. The authors suggested that young adults are better able to manage the coordination and resource distribution of two simultaneously performed tasks. Generally, this finding is in line with the assumption of increasing age differences with more complex tasks (Salthouse, 1992; Voelcker-Rehage et al., 2006) and with the promising character of the dual *n*-back to improve cognitive functioning in older adults.

Effects of dual *n*-back training have by now been studied in young adults only. For example, Jaeggi, Buschkuhl, et al. (2008) found improvements in the reasoning skills of young adults after 12 and more sessions of dual *n*-back training (but see also Redick et al., 2013). Lilienthal, Tamez, Shelton, Myerson, and Hale (2013) reported that after eight sessions of dual *n*-back training the young trainees had improved in a running span WM task, which the authors used to measure the capacity of the focus of attention. We have shown improvements in different executive functions, including WM updating, task switching, and cross-modal attention, following 14 sessions of dual *n*-back training (Salminen, Strobach, & Schubert, 2012). Together these results point toward a potential to improve diverse executive functions by training on the dual *n*-back task. To our knowledge, however, there are no studies to date that have used the dual *n*-back task as a training intervention for older adults.

In the present study, we thus investigated the effects of dual n -back training and transfer to selected executive functions in older adults. We were specifically interested in the question concerning age-related plasticity, and we therefore compared the effects of older adults with the effects of young adults. For this comparison we used the data from a group of young adults who were trained with the same paradigm, and for which we had distinct results at hand (Salminen et al., 2012). To investigate whether possible training effects would transfer to untrained tasks, participants were tested before and after training in tasks engaging executive functions. In the previous study with young adults we specified and systematized different executive functions that corresponded to the requirements of the dual n -back, and we investigated transfer effects from training to tasks measuring these processes separately. The model of executive functions was partially based on the definition by Miyake et al. (2000), which describes WM updating, task switching, and inhibition as core executive functions (see also Miyake & Friedman, 2012). Out of the three functions in this definition, WM updating and task switching promptly correspond to the demands of the dual n -back paradigm. However, we measured a special form of inhibition with a crossmodal attentional blink (AB) paradigm to better match the nature of the dual n -back task. This paradigm requires the inhibition of distractors while allocating attention to two targets presented in two different modalities. Thus, the definition of inhibition is slightly modified from the definition by Miyake et al. (2000) that is specific to the inhibition of prepotent responses rather than to inhibition in attentional processes. These tasks (WM updating, task switching, AB) were used as a platform of the transfer tasks also for the older adults in the current study. A more detailed description of the transfer functions and tasks is presented below.

The first transfer task was designed to investigate whether improvements from WM training with the dual n -back would transfer to another, untrained WM updating task. The n -back task taxes WM updating processes: while new, relevant stimuli have to be coded into WM, old, irrelevant items have to be replaced (Miyake et al., 2000; Morris & Jones, 1990). As the dual n -back requires the processing of both VS and AV stimuli, the WM updating task was designed to also engage these two modalities and therefore it included a VS and an AV task. Both tasks included stimulus sequences of varying lengths, and after each sequence participants had to *reproduce* the four last presented items of the sequence in the correct order (note that the training task rather requires the *recognition* of WM contents). As it cannot be anticipated by the participants at which point the four last items have to be reported, this task requires continuous updating of WM contents. In previous studies older adults improved their WM updating function after several weeks of visual updating training (Dahlin, Nyberg, et al., 2008; Dahlin, Stigsdotter Neely, et al., 2008). In Salminen et al. (2012) the young trainees showed a transfer effect from dual n -back training to the VS WM updating task. Accordingly, we tested whether older adults would show a similar transfer effect from dual n -back training compared with young adults.

In the second transfer task we investigated whether the requirement in the dual n -back task to rapidly switch between the two task streams would transfer to a fast-paced switching task. Task switching typically leads to longer reaction times (RTs) compared with situations in which the same task is repeated in mixed blocks (i.e., switching costs). These costs are explained by the requirement to switch to a new task set and they are associated with task-set reconfiguration processes that need to take place before the execution of the next task (Monsell, 2003; Rogers & Monsell, 1995). Mixing costs are assessed by comparing the (typically longer) RTs in situations in which the same task is repeated in mixed blocks and RTs in single-task blocks (with an isolated presentation of the component tasks). These costs are a measure of sustained control processes in task

switching, including maintaining task-set information and selecting between two tasks. While switching costs are more robust, it is specifically mixing costs that are affected by age (Kray & Lindenberger, 2000) and training (Strobach, Liepelt, Schubert, & Kiesel, 2012). Previous studies have shown that mixing costs can be reduced in older adults by training (Karbach & Kray, 2009). In young adults a transfer effect from dual *n*-back training was not shown to switching costs, but to mixing costs (Salminen et al., 2012). In the present study we investigated whether dual *n*-back training would improve older trainees' task-switching skills as measured by both switch and mixing costs.

Finally, we investigated transfer from dual *n*-back training to inhibition in attentional processes. Successful performance of the dual *n*-back task requires inhibiting irrelevant stimuli during flexible shifts of attention between WM items stemming from two modalities. To test whether the dual *n*-back training improves this skill, we used the AB paradigm. In a typical AB task, two targets are presented in a rapid serial visual presentation (RSVP) stream. When participants have to report both targets and if the second target (Target 2) appears within a time frame of approximately 200–500 ms from the first target (Target 1), its detection/identification is impaired (Chun & Potter, 1995; Dux & Harris, 2007; Jolicoeur, 1998; Shapiro, Raymond, & Arnell, 1994). This effect is generally explained as a depletion of attentional resources by the first target (Target 1), and by inhibitory processes toward distractors, leading to attention “blinking” while Target 2 is presented (Chun & Potter, 1995; Kihara, Yagi, Takeda, & Kawahara, 2011). AB is larger in older than in young adults (Lahar, Isaak, & McArthur, 2001). However, studies have shown that AB can be improved by different training mechanisms in both young (Slagter et al., 2007) and older adults (van Leeuwen, Müller, & Melloni, 2009). In the study by Salminen et al. (2012) an improvement in an AB task following dual *n*-back training was shown in young adults. Therefore, in the present study, we investigated whether there would be a decrease in the magnitude of AB (i.e., AB improvement) in older adults after training on the dual *n*-back task.

Material and methods

Subjects

Altogether 47 older adults were recruited via newspaper announcements and flyers in Berlin and in Munich. They were randomly divided into two groups (note that we applied no matching procedure to avoid systematic allocation of participants to groups). Twenty-six participants (11 male; age range 57–73 years; mean age 65.0 years; one ambidextrous, others right-handed) were assigned to the training program, and 21 participants (eight male; age range 59–77 years; mean age 65.1 years; two left-handed) were assigned to a control group that did not attend training. All older participants were prescreened for cognitive impairments with the Mini-Mental State Examination (MMSE) (Folstein, Folstein, & McHugh, 1975; German version by Kessler, Markowitsch, & Denzler, 2000). All participants scored 28 points or more in the MMSE and thus showed normal cognition. Additionally, the older participants' initial intelligence level was tested with the German Mehrfachwahl-Wortschatz-Intelligenztest (Lehrl, 2005). This test is a multiple-choice vocabulary intelligence test that is generally used to assess premorbid intelligence. There were no significant differences between the training and the control participants. We collected demographic data from the older participants concerning their education level, years of education, and years in occupation, and found no significant differences between the two groups. One participant in the older training group (female, 68 years old) did not

Table 1. Demographic information of the older participants in each group.

Characteristic	Training group ($n = 25$)	Control group ($n = 21$)	p -Value
Age, mean (SD) [range]	64.8 (3.8) [57–73]	65.0 (3.7) [59–77]	.91
Male, %	42.3	40.7	.91
High-school graduate (%)	58.0	59.2	.96
Years of education, mean (SD) [range]	11.4 (1.7) [8–13]	10.9 (1.5) [8–13]	.29
Years of occupation, mean (SD) [range]	39.5 (6.8) [22–51]	35.6 (10.5) [6–48]	.12
MMSE score, mean (SD) [range]	28.35 (5.9) [26–30]	29.11 (1.4) [25–30]	.51
MWT-B IQ, mean (SD) [range]	118.7 (13.5) [97–143]	121.5 (11.3) [95–143]	.42

Notes: SD = standard deviation, MMSE = Mini-Mental State Examination, MWT-B IQ = Mehrfachwahl-Wortschatz-Intelligenztest.

show improvement in the training task in the first seven sessions, reportedly due to a misunderstanding of the task, and this participant was excluded from the analyses. Consequently, the older training group for the final analyses consisted of 25 participants (11 male; age range 57–73 years; mean age 64.8 years; one ambidextrous). Detailed group information of the final sample of the older participants is listed in Table 1.

To investigate age-related plasticity, we compared the results of the older adults to the results of a group of young adults. For this we used the data of the young adults from the study by Salminen et al. (2012). This group of young participants was recruited in Munich and it also represented two groups: a training group of 20 participants (five male; age range 20–31 years; mean age 24.4 years; two left-handed) and a no-training control group of 18 participants (four male; age range 20–32 years; mean age 24.5 years; two left-handed). All participants (old and young) were equally rewarded with a monetary compensation of 8 € per hour, all were native German speakers, and all reported normal or corrected-to-normal vision and normal hearing.

Design and procedure

In the beginning, all participants took part in a pretest session, during which baseline testing for the training task and the transfer tasks (i.e., untrained WM updating tasks, task switching, and AB) were completed. Following this, the training groups attended exactly 14 training sessions spanning over 3–6 weeks (i.e., on average approximately two to five training sessions per week), whereas the control groups underwent no training during this time. Finally, all participants attended a posttest on the training task and the transfer tasks. The period between pretest and posttest was equal between the training and control groups ($M = 30$ and $M = 29$, respectively; $p = .24$). All tasks (training and transfer) were computerized and all tasks (training and transfer) were performed in the laboratories of the departments of psychology in Berlin and Munich (Humboldt-Universität zu Berlin and Ludwig-Maximilians-Universität München). During the training sessions, several participants could complete the tasks in parallel, whereas in the pre- and posttests only one participant was tested at a time; rooms differed between training and transfer sessions. Experimenters were present in all (training and transfer) sessions. In all computerized tasks responses were given on a German standard computer keyboard (QWERTZ) with the exception of task switching in which we used an external response device.

Materials

Training task

The dual n -back training task was a part of the BrainTwister-software package (Buschkuehl et al., 2007) and it consisted of VS and AV stimuli (Figure 1). The VS stimuli were blue squares presented one by one on a black background, appearing randomly in eight possible locations. The AV stimuli were eight German consonants (C, G, H, K, P, Q, T, and W) spoken in a random order via headphones. All stimuli were presented for 500 ms, and the interstimulus interval (ISI) was 2,500 ms, thus resulting in a stimulus presentation rate of three seconds. A central white fixation cross was present continuously. Participants reacted by pressing the key “A” with their left index finger for the VS task (i.e., match of square position in the present and n -back trial) and the key “L” with their right index finger for the AV task (i.e., match of consonant in the present and n -back trial). A new run was commenced by pressing the spacebar. Each run started with instructions about the level of n in the upcoming run and ended with feedback of the participant’s performance in the preceding run. The level of n was always the same in both tasks, with each training session starting from the same level: for the older adults this was n -back level $n = 1$ and for the young adults it was n -back level $n = 2$. A lower level for the older adults compared with the young adults was designed to facilitate their getting started with the task. For each consecutive run, the n -back level was automatically adjusted so that if the participant had at least 90% correct responses in both modalities in the previous run, the level of n in the next run was increased by one. However, if the participant had at most 70% correct responses in either of the modalities, the level of n was decreased by one in the next run, with the minimum level always being $n = 1$. In other cases the n -level stayed constant between successive runs. The older participants completed altogether 30 runs in each training session, whereas for the young participants one training session consisted of 20 runs. The larger amount of runs for the older adults aimed at elevating the intensity of training and therefore to potentially increase the achieved training

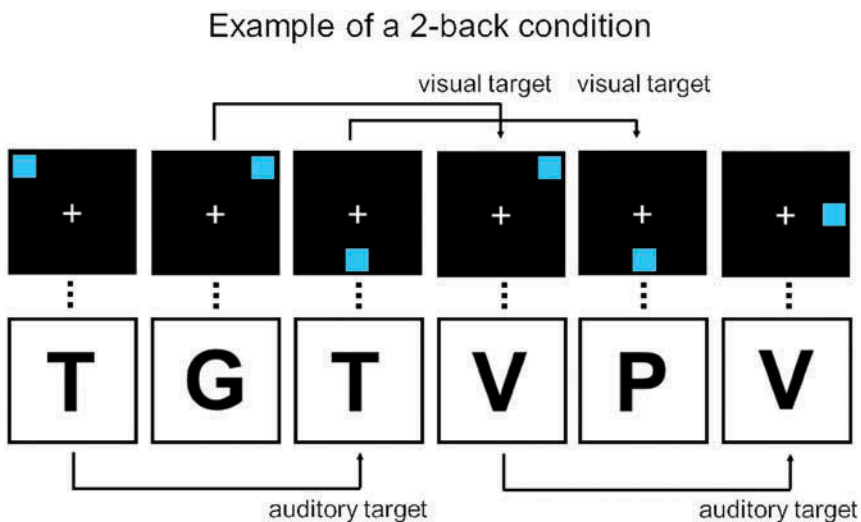


Figure 1. Example of a 2-back condition in the dual n -back task that was used as the training task. The visual and auditory stimuli are presented simultaneously at identical rates. Figure reproduced with permission from Salminen et al. (2012). [To view this figure in color, please see the online version of this journal.]

performance (i.e., achieved level of n) and thus the probability of transfer effects. For all participants, one run consisted of $20 + n$ trials (e.g., a 2-back run contained 22 trials). One training session took approximately 50 min for the older participants and 35 min for the young participants. The dependent measure was the mean n -back level achieved during a training session.

Transfer tasks

WM updating. In this task, the participants performed one block with VS stimuli and another block with AV stimuli. The VS stimuli were black bars that appeared one by one in four different locations on the vertical axis of a computer screen. The AV stimuli consisted of the numbers 1, 2, 3, and 4, spoken in German, and presented through headphones. All stimuli were presented for 2,000 ms with an ISI of 1,000 ms. Each trial included a list of sequentially presented stimuli, and the list lengths were 5, 7, 9, 11, 13, and 15 items. In the VS task, the young participants gave their responses by pressing with their right hand the keys “N” for a bar presented in the uppermost part of the screen, “M” for a bar presented slightly above the middle of the screen, “,” for a bar presented slightly below the middle of the screen, and “.” for a bar presented in the lowermost part of the screen. As this response mapping seemed too challenging for the older participants, their response keys were mapped to reflect the actual position of the bar on the screen, so that they gave their responses on the number pad using the right hand. The older participants thus pressed the key “Num” for a bar presented in the uppermost part of the screen, the “7” key for a bar presented slightly above the middle of the screen, the “4” key for a bar presented slightly below the middle of the screen, and the “1” key for a bar presented in the lowermost part of the screen. On the presentation of the digits 1, 2, 3, and 4 in the AV task, all participants responded by pressing the keys “Y,” “X,” “C,” and “V” with the little, ring, middle, and index fingers of the left hand, respectively. In each block, ten trials were completed. The order of the VS and AV blocks was counterbalanced between participants, and for each individual participant the order was the same in pretest and in posttest. Immediately following the presentation of a list, participants were asked to report the four last presented items of that list in the correct order, starting from fourth last and ending with the last presented item. In each block, participants were instructed to constantly update the four last items during the presentation of the lists. No speeded responses were required, and the next list started automatically after response. Here, the outcome measure was the number of correctly reported 4-item sequences, in each block separately.

Task switching. Each trial consisted of the presentation of a character pair including a digit that was either even (2, 4, 6, 8) or odd (3, 5, 7, 9) and a letter that was either a consonant (G, K, M, R) or a vowel (A, E, I, U). One pair at a time was presented in the center of a cell of a 2×2 grid. The first pair of each block appeared in the upper left cell, and the presentation of the following pairs moved always to the next cell clockwise. Each trial lasted until a participant’s response, or until 5,000 ms had elapsed. The inter-trial interval (ITI) was 150 ms; however, after an erroneous trial it was extended to 1,500 ms and also a tone of 30 ms in length was presented. Participants were instructed to perform a number discrimination task (even vs. odd) and a letter discrimination task (consonant vs. vowel). They were asked to respond as fast and as correctly as possible with a response device including two keys, by pressing the left key with the left index finger for even digits or consonants, and the right key with the right index finger for odd digits or vowels.

Altogether six blocks of 48 trials each were completed. The first two blocks were single-task blocks: one letter categorization and one digit categorization block, and their order was counterbalanced across participants. The last four blocks were mixed blocks, in which both tasks had to be performed so that whenever the stimulus pair appeared in one of the upper cells of the grid, the digit categorization task was to be performed, and whenever the pair appeared in one of the lower cells of the grid, the participant had to perform the letter categorization task. Thus, half of the trials in these blocks were trials in which the same task was repeated from one trial to the next, and the other half were switch trials in which the task switched. After the single-task blocks, the older participants practiced the mixed blocks for 24 trials, in order to get them acquainted with the task. These practice trials were excluded from the analyses. RTs and error rates were used as outcome measures.

Attentional blink. This task included visual and auditory stimuli comprising letters of the alphabet (excluding N, X, C, and Y) and the digits 1, 2, 3, and 4. All visual items appeared sequentially in the same location in the middle of the screen. The auditory stimuli were presented through headphones. Each trial consisted of a concurrently presented visual and auditory stream. The lengths of the streams varied randomly, with one stream including 13, 15, 17, 19, or 21 items. Each stream consisted mainly of letters, except for two digits that appeared concurrently at two positions in the two modalities (i.e., simultaneous visual and auditory digits at position A and simultaneous visual and auditory digits at position B). The positions of the digits in the streams varied randomly, so that the first digits were presented at position 5, 7, 9, 11, or 13 and the second digits followed either three or six positions later. Each stimulus was presented for 80 ms, and with an ISI of 13 ms the presentation rate of the stimuli was 10.75 stimuli per second. Thus, the lag between the first and the second digit pair was either 279 ms or 558 ms. At the end of each trial the participants were asked about the identities of the first visual digit (T1) and of the second auditory digit (T2). Responses were given with the right hand, using the number pad of a keyboard. The first trial of a block was commenced by pressing the spacebar, and the following trials started automatically once the preceding trial had ended. In each trial, first a fixation cross was presented (500 ms), followed by a blank screen (500 ms), after which the auditory and the visual streams started simultaneously. Altogether two blocks with 40 trials each were completed. The first 14 trials in each block served as warm-up trials and they were excluded from the analyses. The outcome measure was the proportion of correctly identified T1 and T2.

Results

Means and standard deviations for each task in pretest and posttest, along with effect sizes (Cohen's *d*) for group-specific pretest–posttest comparisons are presented in [Table 2](#), separately for the older and young training groups and the older and young control groups.

First, a multivariate analysis of variance (MANOVA, Pillai's Trace) was conducted to account for effects of multiple comparisons. This analysis was conducted with Group (training vs. control) and Age (young vs. old) as between-subjects factors and Session (pretest vs. posttest) as a within-subjects factor on the data of each transfer task (i.e., the number of correctly reported items in the WM updating tasks, RTs in each trial type of task switching, and the proportion of correct target identifications in the AB task in each lag; since RTs were our primary measures in task switching, we did not include the error

Table 2. Pre- and posttest performance and the effect sizes (Cohen's *d*) for pre- and posttest comparisons.

Task	Older training group			Older control group			Young training group			Young control group		
	Pretest	Posttest	<i>d</i>	Pretest	Posttest	<i>d</i>	Pretest	Posttest	<i>d</i>	Pretest	Posttest	<i>d</i>
Dual <i>n</i> -back	1.6 (0.3)	2.4 (0.7)	1.49	1.7 (0.4)	1.8 (0.4)	0.25	2.3 (0.4)	4.9 (1.5)	2.37	2.3 (0.5)	2.3 (0.5)	0
WM updating performance in trials correct												
Visuospatial	1.4 (1.5)	2.5 (2.0)	0.62	1.8 (1.5)	2.0 (1.5)	0.13	3.7 (2.2)	5.5 (2.2)	0.82	3.8 (2.7)	3.3 (2.7)	0.19
Auditory-verbal	4.0 (2.2)	4.4 (2.2)	0.18	3.6 (2.6)	3.4 (2.1)	0.08	4.1 (2.3)	6.3 (2.2)	0.98	4.1 (2.4)	5.4 (2.2)	0.56
Task switching RTs in ms/error rates in % correct												
Switch trials	1,918 (366)/ 10.1 (10.7)	1,608 (335)/ 9.9 (10.3)	0.86/0.02	1,922 (367)/ 19.5 (17.4)	1,715 (295)/ 17.5 (17.4)	0.62/0.11	1,348 (279)/ 8.8 (6.7)	1,155 (252)/ 5.6 (4.5)	0.73/0.56	1,418 (225)/ 9.4 (8.0)	1,278 (208)/ 8.3 (6.0)	0.65/0.16
Repetition trials	1,314 (323)/ 4.3 (5.3)	1,118 (230)/ 3.5 (3.4)	0.70/0.18	1,376 (359)/ 11.2 (14.2)	1,185 (293)/ 11.2 (15.4)	0.58/0	877 (190)/ 3.5 (5.3)	722 (141)/ 2.3 (1.7)	0.92/0.30	847 (137)/ 3.3 (6.3)	779 (132)/ 2.9 (3.4)	0.51/0.08
Single-task trials	1,036 (144)/ 3.8 (3.9)	925 (230)/ 3.4 (3.3)	0.58/0.11	997 (192)/ 5.5 (7.0)	977 (249)/ 4.3 (8.6)	0.09/0.15	733 (85)/ 3.9 (2.8)	672 (96)/ 4.1 (3.9)	0.67/0.06	756 (135)/ 6.1 (11.4)	705 (120)/ 3.4 (2.4)	0.40/0.33
Attentional blink in % correct												
T1												
Short lag	77.5 (15.5)	78.8 (15.7)	0.08	66.7 (21.8)	68.0 (21.4)	0.06	85.9 (11.2)	89.8 (9.8)	0.37	76.4 (20.5)	83.8 (11.3)	0.45
Long lag	84.5 (14.2)	80.9 (14.9)	0.25	68.4 (25.4)	71.2 (22.4)	0.12	87.4 (12.8)	91.3 (12.9)	0.30	81.1 (19.4)	89.8 (11.4)	0.55
T2												
Short lag	27.6 (8.1)	28.5 (9.3)	0.10	27.8 (8.7)	26.4 (11.0)	0.14	45.5 (11.0)	56.0 (17.2)	0.73	42.7 (11.6)	44.3 (10.3)	0.15
Long lag	34.1 (9.1)	34.3 (11.2)	0.02	31.3 (13.4)	26.4 (10.3)	0.41	57.7 (16.8)	71.5 (16.4)	0.83	53.5 (20.3)	59.1 (16.9)	0.30

Notes: Data is presented for the training group and the control group in the training and transfer tasks for both age groups separately. Values represent means (and standard deviations).

rate data of these tasks). In this analysis the Group $\times$ Session interaction was significant, $F(9, 70) = 3.05, p < .01, \eta_p^2 = .28$. Further, when we combined all relevant transfer and training task data (i.e., we added the mean achieved level of n in the dual n -back training task) in such a MANOVA, the interaction of Group $\times$ Session $\times$ Age was significant, $F(10, 67) = 3.38, p < .005, \eta_p^2 = .34$. This indicated that there were reliable group-, age-, and task-specific performance changes from pretest to posttest and we thus proceeded to further analyses on each task separately.

In the following we report separate follow-up analyses for the tasks. The dual n -back training task as well as the VS and AV WM updating transfer tasks were each submitted to three-way mixed-design ANOVAs with the between-subjects factors Group (training vs. control) and Age (old vs. young) and with the within-subjects factor Session (pretest vs. posttest). Similar ANOVAs were conducted for task switching and the AB task with one added factor for each task. In task switching, the additional within-subjects factor Trial type was included in the analyses (repetition trials vs. single-task trials in the analysis for mixing costs, and switch trials vs. repetition trials in the analysis for switching costs). In the AB task the additional within-subject factor Lag (short vs. long) was included. As our interest was to compare training and transfer effects between the two age groups, for the sake of clarity we mainly report interactions including combinations of Group and Session (e.g., Group $\times$ Session or Group $\times$ Age $\times$ Session). In case of significant interactions, follow-up analyses are reported.

Training task

In the 2 (Group) $\times$ 2 (Age) $\times$ 2 (Session) ANOVA the interaction Group $\times$ Age $\times$ Session reached significance [$F(1, 80) = 29.94, p < .001, \eta_p^2 = .27$], indicating that the training group's improvement over the control group was larger in the young participants than in the older participants (Figure 2). Pairwise t -tests conducted on the mean achieved n -back levels in pre- and posttests showed significant differences in performance between the two sessions for both the older trainees [$t(24) = -7.04, p < .001$, Cohen's $d = 1.58$] and the young trainees [$t(19) = -8.70, p < .001$, Cohen's $d = 2.35$]. This confirms that both training groups improved in the training task from pre- to posttest, whereas no such improvement was observed in either of the control groups ($p = .08$ for the older participants and $p = .44$ for the young participants). When comparing the pretest performance of the training group to the control group, there was no significant difference in the mean achieved n -back level for either of the age groups ($p = .68$ for older trainees vs. older controls and $p = .66$ for young trainees vs. young controls). However, in posttest the older trainees' performance significantly differed from the older controls' performance [$t(39) = 3.82, p < .001$, Cohen's $d = 1.11$], a finding that was similar in the group of young participants [$t(24) = 7.15, p < .001$, Cohen's $d = 2.29$]. Interestingly, when comparing the posttest performance of the older trainees to the pretest performance of the young trainees, there was no significant difference observed ($p > .05$). However, in pretest the older trainees performed on a significantly lower level than the young trainees ($M = 1.64$ and $M = 2.34$, respectively), [$t(43) = -7.24, p < .001$, Cohen's $d = 2.14$]. That is, after training the performance difference between older adults and untrained young adults had disappeared.

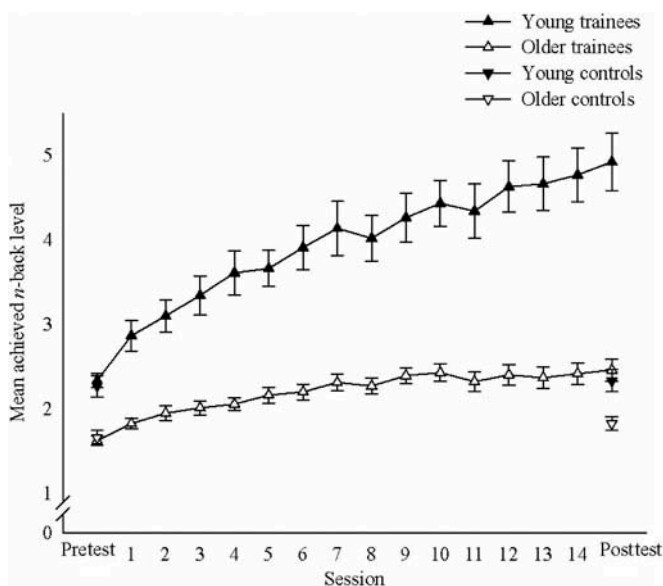


Figure 2. Improvement in the performance of both of the training groups through the training period and the performance of the control groups in the pre- and posttests in the dual n -back task. For each session, the mean n -back level is presented. Error bars indicate standard errors of the mean.

Transfer tasks

VS WM updating task

The 2 (Group) $\times$ 2 (Age) $\times$ 2 (Session) mixed-design ANOVA yielded a significant interaction between Group and Session [$F(1, 80) = 9.28, p < .05, \eta_p^2 = .10$], showing that the training groups improved significantly from pretest to posttest ($M = 1.44$) whereas the control groups did not ($M = -0.10$) (Figure 3). This training gain was similar for both age groups, as the interaction Group $\times$ Age $\times$ Session did not reach significance ($p = .17$). Follow-up comparisons confirmed that both the older trainees [$t(24) = -2.96, p < .01$, Cohen's $d = 0.62$] and the young trainees [$t(19) = -2.49, p < .05$, Cohen's $d = 0.82$] showed a significant improvement from pretest to posttest, whereas no change was observed in the control groups ($p = .40$ for the older participants and $p = .48$ for the young participants). Again, we compared the posttest performance of the older trainees to the pretest performance of the young trainees. No significant difference was found ($p > .05$). A pretest comparison between the two groups did, however, reveal a significant difference [$t(43) = -4.16, p < .001$, Cohen's $d = 1.22$] such that the young trainees outperformed the older trainees at pretest ($M = 3.70$ and $M = 1.40$, respectively). That is, similar to the training task, the older trainees reached after training the baseline performance level of the young participants in the VS WM updating task.

AV WM updating task

There was no interaction either of Group $\times$ Session ($p = .38$) or of Group $\times$ Age $\times$ Session ($p = .84$) (Figure 4). Thus, there were no training-related improvements in either of the age groups.

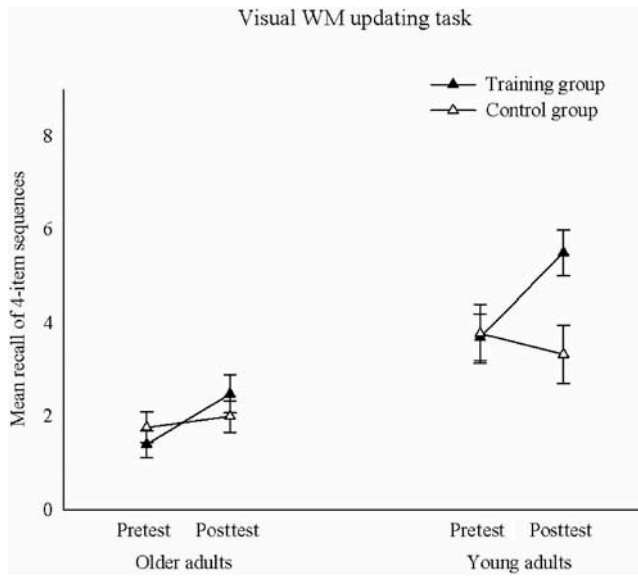


Figure 3. The number of correctly reported four-item sequences in the visuospatial (VS) WM updating task. Performance for all groups is illustrated separately for pretest and posttest. Error bars indicate standard errors of the mean.

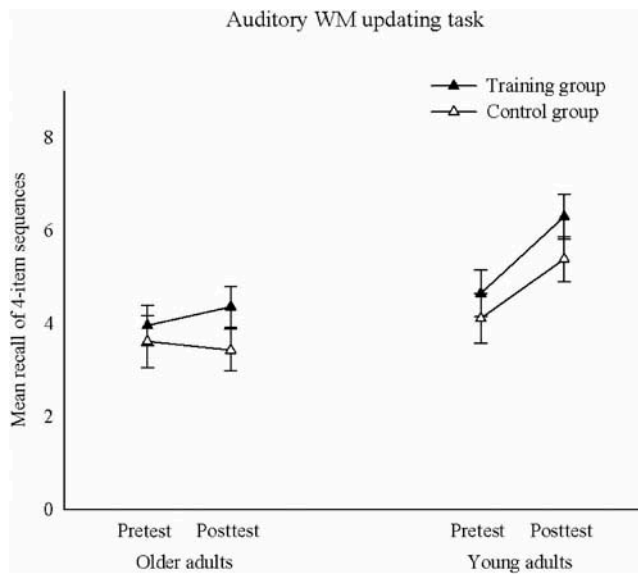


Figure 4. The number of correctly reported four-item sequences in the auditory-verbal (AV) WM updating task. Performance for all groups is illustrated separately for pretest and posttest. Error bars indicate standard errors of the mean.

Task switching

One participant of the young trainees was excluded from the analyses due to a high percentage of errors in all trial types in the posttest (> 87%). In the RT analyses, trials with incorrect responses (11.6% of trials in the old group and 5.6% of trials in the young group) were excluded.

Mixing costs. The mixed-design ANOVA on RTs yielded a significant interaction between the factors Group, Age, Session, and Trial type [$F(1, 79) = 5.04, p < .05, \eta_p^2 = .06$]. That is, there were specific effects of training on repetition and single-task trials, and these effects were different between older and young trainees and control participants. Therefore, we conducted 2 (Group) $\times$ 2 (Session) $\times$ 2 (Trial type) mixed-design ANOVAs separately for the two age groups. For the older participants, the main effect of Session [$F(1, 44) = 45.14, p < .001, \eta_p^2 = .51$] reached significance, showing that all older participants were faster in posttest ($M = 1,051$ ms) than in pretest ($M = 1,181$ ms). As the Group $\times$ Session interaction was not significant ($p = .22$), the performance improvement was not specific to training. For the young participants, the analysis of mixing costs also revealed a main effect of Session [$F(1, 35) = 51.14, p < .001, \eta_p^2 = .59$], indicating faster RTs in posttest ($M = 719$ ms) than in pretest ($M = 803$ ms). Here, the Group $\times$ Session interaction reached significance [$F(1, 35) = 4.38, p < .05, \eta_p^2 = .11$], indicating that the young training group improved from pretest to posttest more ($M = 108$ ms) than the young control group ($M = 59$ ms). Additionally, the Session $\times$ Trial type $\times$ Group interaction reached significance [$F(1, 35) = 4.55, p < .05, \eta_p^2 = .12$], showing that the training-related improvement was trial-type specific. We therefore conducted two further Group $\times$ Session ANOVAs separately on the RTs in single-task trials and in repetition trials. The interaction between the factors Group and Session was significant for the repetition trials [$F(1, 35) = 8.52, p < .01, \eta_p^2 = .20$] but not for the single-task trials ($p = .73$). This indicated that only in repetition trials and not in single-task trials the young adults' performance was improved by training. As a result, the improvement (i.e., reduction) in the mixing costs from pretest to posttest was larger in the training group ($M = 95$ ms) than in the control group ($M = 17$ ms). Taken together, a transfer effect to mixing costs was only found for the young trainees, whereas the older participants showed no such transfer.

A similar four-way mixed-design ANOVA was conducted for the error rates. There were no significant interactions with factor combinations of Group $\times$ Session (all $ps > .20$). Thus, there were no training-related effects on the error rates in either of the age groups.

Switching costs. There were no training-related improvements in the 2 (Group) $\times$ 2 (Age) $\times$ 2 (Session) $\times$ 2 (Trial type) mixed-design ANOVAs on RTs and error rates in the switching costs of either of the age groups, as indicated by non-significant interactions in factor combinations of Group $\times$ Session (all $ps > .11$).

Attentional blink

Due to technical problems, the data of one older male participant from the training group was not recorded correctly and thus this participant was excluded from the analyses.

Target 1. The mixed-design ANOVA on the percentage of correct Target 1 reports yielded no training-related improvement in either of the age groups since there was no significant interaction in factor combinations of Group $\times$ Session (all $ps > .15$).

Target 2. Analyses were conducted only for the trials, in which Target 1 was identified correctly. There was a significant interaction between the factors Group and Session [$F(1, 79) = 7.65, p < .01, \eta_p^2 = .09$], which shows that the training groups improved more from pretest to posttest than the control groups did ($M = 6.4\%$ and $M = 0.3\%$, respectively). This effect was similar for both age groups, as the Group $\times$ Age $\times$ Session interaction did not reach significance ($p = .28$). However, a more detailed analysis revealed that there was a training-related improvement in T2 identification (over both lags) only for the young adults [$F(1, 36) = 6.14, p < .05, \eta_p^2 = .15$]. For the older adults, this interaction was driven by an artifact of the control group showing a numerical impairment from pretest to posttest, whereas the training group showed no change between the sessions (Table 2).

Discussion

The goal of this study was to investigate whether executive functions can be improved with WM training in older adults. To date, studies approaching this issue have been scarce (but see Brehmer et al., 2012; Dahlin, Stigsdotter Neely, et al., 2008; Zinke et al., 2014), although several studies have shown transfer effects after WM training to other cognitive functions, including verbal learning and memory (Richmond et al., 2011), language comprehension (Carretti et al., 2012), and VS skills (Stepankova et al., 2014). Executive functions, however, are essential in successful autonomous functioning in everyday life (Miyake & Friedman, 2012) and they are generally negatively affected by aging (Raz, 2000; Salthouse et al., 2003). Hence, it would be of importance to gain more insight into the potentials to compensate for age-related decline in these functions.

In the present study, we trained older adults on the dual n -back task, which is a process-based training paradigm developed to improve WM processes generally and irrespective of the trained task and material (Buschkuhl et al., 2007). The dual n -back task consists of a VS and an AV n -back task presented in parallel. This dual-task component distinguishes it from training interventions previously studied in older adults as they have used WM training tapping only one modality at once (visual or auditory). In our view, a more complex training paradigm would be more apt to balance out adult age differences, as studies have shown age differences to be more prominent in more complex tasks (Jaeggi, Schmid et al., 2008; Salthouse, 1992; Voelcker-Rehage et al., 2006). In a previous study with young adults we found transfer effects after dual n -back training to several executive functions including WM updating, task switching, and inhibition in attentional processes (Salminen et al., 2012). Therefore, in the present study we investigated specifically the gain in the training task as well as transfer effects in older adults, and compared these effects with our previous findings in young adults.

Training task

Both older and young adults improved in the dual n -back task across 14 sessions of training, compared with their respective control groups that did not undergo any training and that showed no improvements in their performance from pretest to posttest. The magnitude of the training effect was different between the two age groups: young adults showed a more prominent improvement than older adults. These results are in accordance

with findings from other studies that have shown specific improvements after WM training in older adults (Borella, Carretti, Riboldi, & De Beni, 2010; Buschkuhl et al., 2008; Li et al., 2008) as well as with findings that show larger training effects in young adults than in older adults (Brehmer et al., 2012; Dahlin, Nyberg, et al., 2008; Dahlin, Stigsdotter Neely, et al., 2008). However, the current study is the first to show such effects following training on a complex WM task including a dual-task component.

Interestingly, when comparing the posttest performance of the older trainees to the pretest performance of the young trainees we found no significant differences. That is, after approximately three weeks of WM training, the older adults had reached the performance level of the untrained young adults in the dual n -back task. This finding coincides with the observations from Li et al. (2008), whose participants trained on two VS n -back tasks over the course of 45 days and after training the older participants performed on a similar level as the young adults before training. In the present study the dual n -back training task required simultaneous processing of two different modalities and it was therefore more complex than the training tasks implemented by Li et al. (2008). Additionally, we show that the older trainees can progress to the baseline level of young adults with a much shorter training period than in the study by Li et al. (2008) (14 vs. 45 training sessions). Hence, we have shown that even with such a complex training task the performance differences between young and older adults can be balanced out with only 14 sessions of training.

Transfer tasks

An important question in the training literature is whether the achieved improvements after training are task-specific, or whether the training effects generalize also to improvements in untrained tasks and functions. We tested for such transfer effects with three tasks engaging different executive functions that we considered to be related to the performance requirements of the dual n -back task. These functions were adapted to partially match the executive functions specified by Miyake et al. (2000): 1) updating of information in WM and replacing old, irrelevant information with new, relevant ones (in the context of the WM updating task), 2) selecting relevant task information, and flexible switching between two tasks (task-switching paradigm), and 3) inhibition; however, in the current study we modified the operationalization of inhibition to match the requirements of the dual n -back task more precisely and, thus, our third transfer function measured the inhibition of a stimulus from one modality while allocating attention to a stimulus from another modality (in the AB paradigm).

We found an improvement of the older trainees in the VS WM updating task in which a similar training-related improvement was found in a previous study with young adults (Salminen et al., 2012; see Karbach, Strobach, & Schubert, *in press*, for similar effects in children). This effect was significant in both age groups compared to an untrained control group. Out of all the current transfer tasks the WM updating task was most similar to the training task. However, it differs from the training task in several crucial aspects, which still points to process-based rather than task-specific and potentially strategic improvements after training. First of all, the dual n -back and the WM updating task require different recalling processes: while in the dual n -back participants have to *recognize* whether a currently presented stimulus matches a stimulus presented n steps back, in the WM updating task the four last presented items have to be *retrieved* from WM. Second, the stimulus material and instructed responses are different in the two tasks (see Methods). These differences between the training and the transfer task point strongly

toward an improvement in the underlying WM processes rather than to an improvement specific to the trained material. Additionally, as in the training task, also in this transfer task the older trainees reached the baseline level of the young adults after the training period. This finding thus provides very encouraging evidence concerning the possibilities of compensating for age-related cognitive decline beyond the trained task and material.

The older adults did not show transfer to tasks measuring other executive functions. One explanation for the lack of transfer effects was provided by Dahlin, Nyberg, et al. (2008) and Dahlin, Stigsdotter Neely, et al. (2008) who speculated that the determining factor for transfer is not the training gain as such but the achieved final level of performance after training. According to these authors the achieved level of performance after training reflects the efficiency of training. One could assume that in the present study the older trainees did not reach a high enough level at the end of training in order for the training effects to transfer more generally. The encouraging finding of transfer to another WM task (VS WM updating) implies that a task- and material-independent improvement had occurred. Along these lines, it is possible that the older adults would require a longer period of training than the young adults to show broader transfer effects. Also, Buschkuehl et al. (2008) and Li et al. (2008) found transfer only to another task tapping the same function as the trained task. However, the present study is the first to indicate that even with a complex WM training paradigm that also includes the involvement of executive control functions, broad transfer effects are challenging to demonstrate.

Further interpretations for the lack of transfer effects are rather specific to the transfer tasks. In the WM updating task we found transfer only in the VS but not in the AV task in both young and older trainees. As already proposed in Salminen et al. (2012), there are diverse possibilities for the absence of transfer in the AV version. Auditory WM functions are perhaps more automatized than visual WM functions, considering the requirements to understand speech, for example (Baddeley, 2003). Therefore, since the level of difficulty (i.e., the level of n) was always identical in the VS and the AV n -back task, it is possible that the auditory WM processes were not strained enough to allow transfer to an untrained task with untrained stimuli. Moreover, visual WM functions have been shown to be more closely related to executive control processes (Miyake, Friedman, Rettinger, Shah, & Hegarty, 2001). Transfer to the VS WM updating task was thus presumably possible due to training-related improvements in these more central executive processes.

In task switching we found group- and age-specific effects on mixing costs. An analysis made separately for the two age groups revealed a training-related improvement in young but not in older adults. For neither of the age groups were there improvements found in the switching costs. Thus, although other studies have shown that specific task-switching training with young and older adults can lead to improvements in task-switching skills that exceed the retest effects of no-training control groups (Karch & Kray, 2009; Kray & Lindenberger, 2000), with the WM training paradigm used in the current study such a training-related improvement was only observed in the mixing costs of young adults.

As for the AB task, we found no training-related improvement in the older participants; this finding is in contrast to the findings with young adults, who showed an improvement in T2 detection after dual n -back training (Salminen et al., 2012). One explanation for these different transfer effects may result from the general magnitude of the AB effects in both age groups: the older adults showed a generally larger AB effect than the young participants, as they identified T2 only in approximately 29% of the trials correctly, whereas for young adults this rate was 54%. The lack of transfer in older adults might therefore have been due to the high demands of the crossmodal AB transfer task. A

previous study reported a reduced AB in older adults following meditation training (van Leeuwen et al., 2009), but the current results support the findings from a study by Basak, Boot, Voss, and Kramer (2008), in which video game training did not affect the magnitude of AB in older adults.

Limitations

The research field concerning cognitive plasticity is still relatively young, and studies on cognitive training are affected by several restrictive factors. For example, the use of no-contact control groups in training studies has been a common target for criticism (Green, Strobach, & Schubert, 2014; Schubert & Strobach, 2012). It is important to bear in mind that when comparing the improvements of a training group to a no-contact control group, unspecific motivational factors induced by the sole active participation on an intervention are likely to play a role in the training and transfer effects. However, such unspecific effects would be expected to be present across the whole battery of tasks that the participants complete. In the present study we, however, observed a differential picture with a significant effect in the training task and a transfer effect to another WM task that functionally resembled the training task, whereas no training-related effects were observed in the other tasks. Additionally, recent studies that have compared active and passive control groups have not found significant differences between the groups with regard to transfer effects (Au et al., 2014; Karbach & Verhaeghen, 2014).

One could also claim that our observed transfer effect to the VS WM updating task was a result of the participants having perceived the VS WM updating task as most similar to the training task, and consequently, this could have led to more motivation to perform best at this task. However, we are not convinced that this is a valid argument since such an expectation-based assumption requires several conditions to be met (e.g., subjects need to be aware of the study's hypotheses and they would also need an explicit understanding of how the hypotheses should be expressed in the data; Green et al., 2014). Since such assumptions are more than speculative, we are confident that the specific effects observed after training can be attributed to training per se and not just on (active) participation in a scientific study. In addition, in the current study we have compared the findings of an experimental group with those of a passive control group in both age groups, which should have left the critical conclusions about age-specific training effects unchanged.

A specific concern in training studies that compare the effects and efficiency of training between young and older adults is general performance differences between the two age groups (see Table 2). In the present study, the older participants had often difficulties in performing the transfer tasks, which would manifest as not performing as instructed (e.g., in the WM updating task older participants would sometimes give their responses in a reverse order from last to first, although instructed contrarily) or as responding randomly (in the AB task some participants performed below the probability of random guessing when reporting T2). Accordingly, limited transfer effects with the older adults in the present study might, at least partially, be due to the transfer tasks being overwhelmingly challenging. In future studies the transfer tasks should therefore be better adapted to the performance level of the age group so that true changes in performance can be measured convincingly.

Another concern specific to the current study might relate to the possible effects of fatigue during the training sessions. As clarified in the Methods section, we chose longer training sessions for the older trainees than for the young trainees (30 vs. 20 runs per

session) in order to intensify training and consequently to increase the probability of transfer effects. However, although the training sessions were self-paced (each run was commenced by the participant, and therefore the participants could take small breaks between the runs), it is possible that the longer training sessions induced within-session fatigue. We tested for this possibility by comparing the older trainees' within-session performance (mean achieved n -back level) for runs 1–20 with runs 1–30 in each session. There were no significant differences found (all $ps > .23$). We also compared the older trainees' performance in runs 21–25 with runs 26–30 in each session, and we found no significant differences (all $ps > .38$). Thus, it seems that although the older participants did not reach significantly higher levels of n in the last runs of each session, they also did not show a decrease in performance and consequently no evidence of within-session fatigue.

Conclusions

Studying cognitive plasticity across the life-span has become more and more important as both life expectancy and the proportion of elderly in the population are increasing steadily. This trend has thus inspired studies investigating the possibilities to compensate for age-related decline and to delay or even to hinder the development of age-related cognitive pathologies. The present study provides encouraging evidence toward cognitive plasticity still at an older age, shown as balancing out the performance differences to young adults by training and as a reliable improvement in a trained and an untrained WM task.

Disclosure statement

No potential conflict of interest was reported by the authors.

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ORCID

Tiina Salminen  <http://orcid.org/0000-0002-3920-2182>

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